BIO 005 HUMAN PHYSIOLOGY · INSTRUCTOR ONLY

# Clicker key: Week 9, The Heart as a Pump

Three class sets, picked on the deck's title bar. Concept clickers: 30 s, confidence points. Label-it clickers: 15 s, one tap. Rooms HEART9, HEART9B, HEART9C on the dashboard.

| SET A · ROOM HEART9 · KEY: BCADADBCDBAC | | | ANSWER | |
|---|---|---|---|---|
| 1 | CONCEPT | A drug blocks myosin light chain kinase in vascular smooth muscle. What happens to the vessel, and why? | B | It relaxes, because myosin cannot be phosphorylated and cannot cycle |
| 2 | LABEL | Name the structure at the pin. | C | Phosphorylated myosin light chain |
| 3 | CONCEPT | A toxin destroys the gap junctions in the intercalated discs but leaves the desmosomes intact. What happens to the heartbeat? | A | The cells stay attached but no longer fire together, so the myocardium stops acting as one unit |
| 4 | CONCEPT | A patient has a tumor of the adrenal medulla that dumps catecholamines into the blood in bursts. Why do the episodes last many minutes, when a sympathetic nerve response is over in milliseconds? | D | Hormones in blood are cleared slowly by the liver; nerve-released norepinephrine is taken back up at once |
| 5 | CONCEPT | A drug blocks HCN (funny current) channels in the SA node. What changes on the pacemaker potential, and what happens to heart rate? | A | The slow depolarization is shallower, threshold takes longer to reach; rate falls |
| 6 | LABEL | Name the structure at the pin. | D | Pacemaker potential (slow depolarization) |
| 7 | LABEL | Name the structure at the pin. | B | Phase 3: repolarization (ventricle) |
| 8 | CONCEPT | On a strip there are 25 small boxes between R waves, and every P is followed by a QRS with a PR of 0.16 s. Rate and rhythm? | C | 60 a minute, sinus rhythm |
| 9 | LABEL | Name the structure at the pin. | D | T wave |
| 10 | CONCEPT | During which phase are all four valves closed and ventricular pressure rising? | B | Isovolumic contraction |
| 11 | LABEL | Name the structure at the pin. | A | Isovolumic relaxation |
| 12 | CONCEPT | EDV 150 mL, ESV 90 mL, HR 80. Stroke volume, ejection fraction and cardiac output? | C | 60 mL, 40%, 4.8 L/min |

| SET B · ROOM HEART9B · KEY: DBCACDBADBCA | | | ANSWER | |
|---|---|---|---|---|
| 1 | CONCEPT | A tissue's smooth muscle cells all contract together even though only a few of them receive a nerve ending. Which kind of smooth muscle is this, and what makes it possible? | D | Single-unit; gap junctions pass the signal cell to cell |
| 2 | CONCEPT | Which pairing of muscle type and calcium source is correct? | B | Cardiac: extracellular trigger $Ca^{2+}$ then SR release |
| 3 | LABEL | Name the structure at the pin. | C | Intercalated disc |
| 4 | CONCEPT | Why is the adrenal medulla called a modified sympathetic ganglion? | A | Its chromaffin cells are postganglionic neurons without axons, driven by preganglionic cholinergic neurons |
| 5 | CONCEPT | A ventricular cell is bathed in a solution with no $Ca^{2+}$. Which phase of its action potential disappears first, and what happens to the twitch? | C | Phase 2, the plateau; the twitch fails |
| 6 | LABEL | Name the structure at the pin. | D | Threshold in the SA node |
| 7 | CONCEPT | A strip shows P waves at 80 a minute and wide QRS complexes at 35 a minute with no fixed relationship between them. What is going on? | B | Complete heart block; the ventricles are on a Purkinje escape rhythm |

| SET B · ROOM HEART9B · KEY: DBCACDBADBCA | | | ANSWER | |
|---|---|---|---|---|
| 8 | LABEL | Name the structure at the pin. | A | Sinoatrial (SA) node |
| 9 | LABEL | Name the structure at the pin. | D | ST segment |
| 10 | CONCEPT | A murmur is heard between S1 and S2, loudest at the apex. Which valve problem fits? | B | Mitral regurgitation |
| 11 | LABEL | Name the structure at the pin. | C | Pulmonic area |
| 12 | CONCEPT | A patient's venous return rises suddenly when they lie down. Without any nerve input, what happens to the next stroke volume and why? | A | It rises, because greater EDV stretches the fibers and they contract harder (Frank–Starling) |

| SET C · ROOM HEART9C · KEY: DBCABADCDBCA | | | ANSWER | |
|---|---|---|---|---|
| 1 | CONCEPT | Smooth muscle in the bladder wall holds tension for an hour using very little ATP. Which feature explains that? | D | The latch state: crossbridges stay attached after the light chain is dephosphorylated |
| 2 | LABEL | Name the structure at the pin. | B | Sarcoplasmic reticulum (smooth muscle) |
| 3 | CONCEPT | A student says the heart could be tetanized if you stimulated it fast enough, like skeletal muscle. What is the best correction? | C | Wrong; the cardiac refractory period lasts almost as long as the twitch, so a second contraction cannot stack on the first |
| 4 | LABEL | Name the structure at the pin. | A | Ryanodine receptor (cardiac SR) |
| 5 | CONCEPT | Skeletal muscle arterioles dilate during a fight or flight response even though they have almost no sympathetic nerve supply. Which signal reaches them? | B | Epinephrine from the adrenal medulla acting on $\beta_2$ receptors |
| 6 | CONCEPT | Which statement correctly contrasts the pacemaker cell with the contractile cell? | A | Pacemaker upstroke is $Ca^{2+}$ and there is no stable resting potential; contractile upstroke is $Na^{+}$ from a stable −90 mV |
| 7 | LABEL | Name the structure at the pin. | D | Phase 0: rapid upstroke |
| 8 | CONCEPT | Suppose a drug removed the AV node delay. What would go wrong with the pump? | C | The ventricles would contract while the atria are still emptying into them, so ventricular filling drops |
| 9 | LABEL | Name the structure at the pin. | D | Purkinje fibers |
| 10 | CONCEPT | On a pressure volume loop, afterload rises (the patient's arterial pressure is high). How does the loop change? | B | Taller and narrower: point B is higher and less blood is ejected |
| 11 | LABEL | Name the structure at the pin. | C | Point B: aortic valve opens |
| 12 | CONCEPT | A $\beta_1$ blocker is given. Predict the effect on heart rate and contractility, and name the step that is interrupted. | A | Both fall; norepinephrine can no longer raise cAMP in SA node and contractile cells |